6. It has a half life of approximately 5 days.

iv) IgD

1. IgD accounts for about less than 1% of the total immunoglobulin pool.
2. One unique structural feature is the presence of only a single H-H inter chain bond along with two H-L interchain bonds.
3. It has a monomeric structure similar to that of IgG.
4. IgD antibodies are abundant in combination with IgM on the surface of B cells and thus are part of the B cell receptor complex. Therefore their function is to signal the B cell to start antibody production upon initial antigen binding.
5. It has a half life of 2-3 days.

v) IgE

1. IgE accounts for only 0.004% of serum immunoglobulin. It has a monomeric structure. It is also called reagin or reaginic antibody.
2. The skin sensitizing and anaphylactic antibodies belong to this class.
3. The Fc portion of IgE can bind to Fc receptors specific for IgE that are found on mast cells, eosinophils and basophils. Thus these cells can become coated with IgE molecules. When two cell- bound IgE molecules are cross linked by binding to the same antigen, the cells degranulate. This degranulation releases histamine and other mediators of inflammation.
4. IgE also stimulates production of an excessive number of eosinophils in the blood (eosinophilia) and increased rate of movement of the intestinal contents (gut hypermotility) which aid in the elimination of helminthic parasites. IgE has a half life of 2-3 days.



Polyclonal Antibody: A mixture of antibodies produced by a variety of B-cell clones that have recognized the same antigen. Although all of the antibodies react with the immunizing antigen, they differ from each other in amino acid sequence.

Breast Milk: Breast milk is uniquely suited to the human infant's nutritional needs and is a live substance with unparalleled immunological and anti-inflammatory properties that protect against a host of illnesses and diseases for both mothers and children. All five classes of immunoglobulins have been found in human milk, but by far the most abundant type is IgA, specifically the form known as secretory IgA.

Antitetanus Serum: Antitetanus serum, also known as tetanus immune globulin (TIG) is made up of antibodies against the tetanus toxin. It is used to prevent tetanus in those who have a wound that is at high risk and have not been fully vaccinated with tetanus toxoid. It is also used to treat tetanus along with antibiotics and muscle relaxants. It is given by injection into a muscle.

TN_GOVT_XI_Micro_Biology_CH13.indd 243 16-02-2019 10.36.11 AM